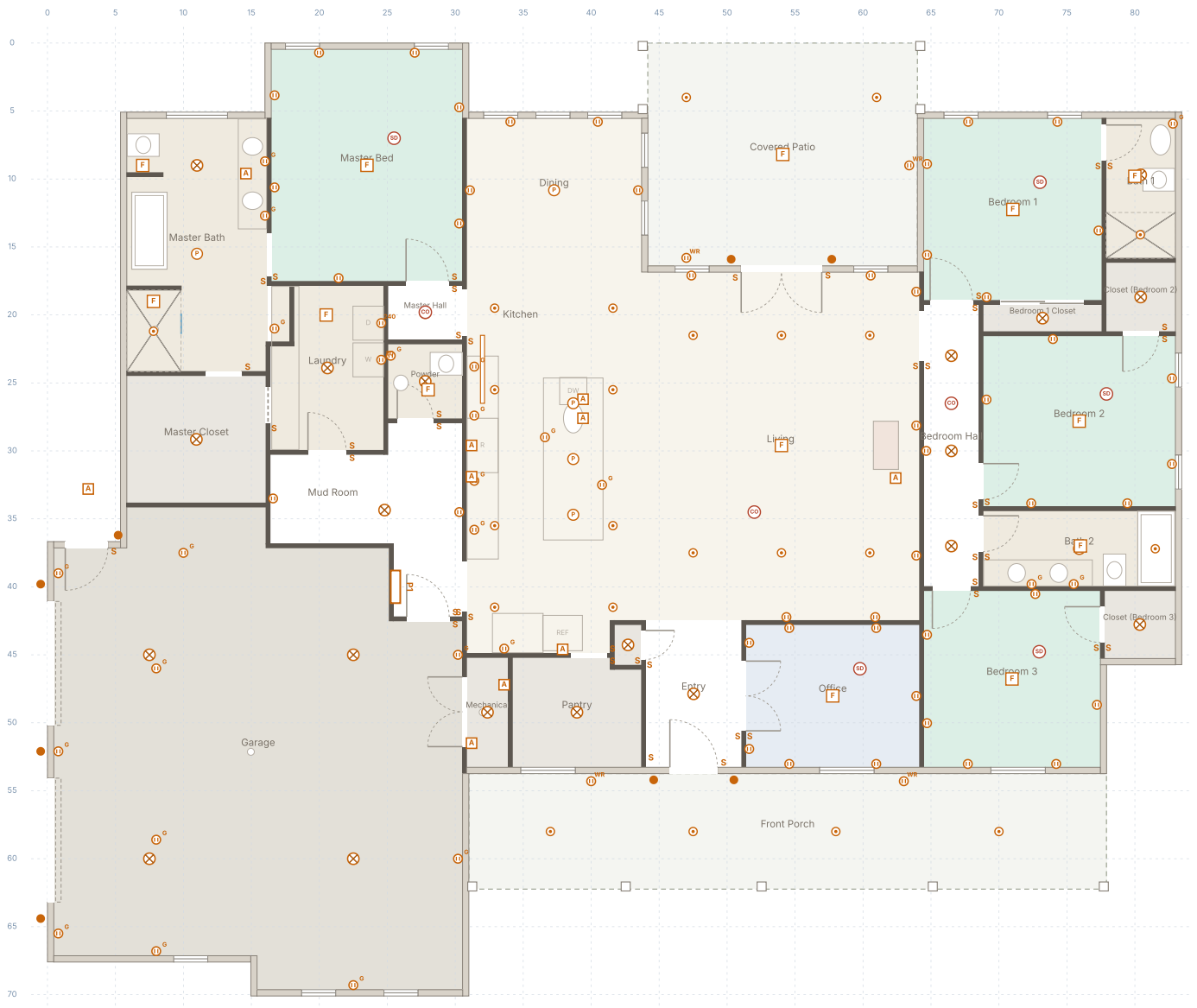
REAR · PATIO



FRONT · PORCH

Receptacles R, switches S, lights L, fans F, appliances A, alarms D. · Pick a device for its circuit; pick a circuit row to light up everything on it. · Main panel P1 in the garage. Calculated load 134 A; 200 A service.

draft

E-1 Electrical · 64 Stone Sheep Circle

First-draft device layout to NEC 2023 minimums, for the electrician to lay out and the owner to walk through. App version v3. From the owner's plans (Larson House, 6/11/26).

Notes

1. Main panel P1, 200 A, in the garage on the wall to the mud room (W43), 36 spaces used of a 40-space panel. Penetrations through the garage wall fire-caulked.
2. Service size from the NEC 220.82 calculation below: about 134 A with a heat pump assumed; 200 A leaves room for an EV charger (add a 50 A 240 V circuit in the garage if wanted).
3. AFCI on every 120 V 15/20 A branch circuit in the living spaces (210.12); GFCI in baths, kitchen counters, laundry, garage, outdoors, dishwasher (210.8).
4. Tamper-resistant receptacles throughout (406.12). Receptacles at 16" to center, counters 44", vanity 40".
5. Smoke alarms in every bedroom, outside the sleeping areas and on the living level; combination smoke/CO in the halls outside the sleeping areas (attached garage and wood fireplace); all interconnected.
6. Kitchen and living ceilings are vaulted 4:12: sloped-ceiling recessed housings and a fan downrod sized for the slope.
7. Device positions are drawn to the room, not measured to studs: the electrician sets final boxes at the walk-through.

Open items

- Range fuel: the 48" range (BS655Z-48) could be gas, dual-fuel or electric. Electric is assumed (50 A 240 V); a gas or dual-fuel range drops that circuit and adds a gas line.
- Heating and water heating fuel (heat pump / gas furnace / electric) set the mechanical circuits and the service load.
- Meter and service entrance location (overhead or underground, which side of the house) is not on the site plan.
- EV charger in the garage: yes or no.
- Lighting choices (cans vs. surface fixtures, pendants, fan locations) for the owner's walk-through.
- Low-voltage: network drops, TV at the fireplace wall, doorbell, security.

Circuit schedule, main panel P1

Ckt	Serves	Breaker	Poles	Protection	Wire (Cu)	Devices	Notes
1	Range	50 A	2-pole	240 V	6 AWG	1	electric range assumed; drop if gas
2	Dishwasher	20 A	1-pole	GFCI	12 AWG	1	—
3	Disposal	20 A	1-pole	GFCI	12 AWG	1	—
4	Refrigerator	20 A	1-pole	—	12 AWG	1	—
5	Range hood	15 A	1-pole	—	14 AWG	1	—
6	Air handler / furnace	30 A	2-pole	240 V	10 AWG	1	size to the equipment; electric strip heat would need 60 A
7	Water heater	30 A	2-pole	240 V	10 AWG	1	heat-pump or electric; drop if gas

Ckt	Serves	Breaker	Poles	Protection	Wire (Cu)	Devices	Notes
8	Heat pump / AC	30 A	2-pole	240 V	10 AWG	1	size to the equipment nameplate
9	Fireplace blower	15 A	1-pole	AFCI	14 AWG	1	—
10	Kitchen counter 1 (small appliance)	20 A	1-pole	GFCI/AFCI	12 AWG	4	—
11	Kitchen counter 2 (small appliance)	20 A	1-pole	GFCI/AFCI	12 AWG	3	—
12	Washer	20 A	1-pole	GFCI/AFCI	12 AWG	1	laundry circuit
13	Dryer	30 A	2-pole	240 V	10 AWG	1	—
14	Laundry counter	20 A	1-pole	GFCI/AFCI	12 AWG	1	—
15	Bath receptacles (all baths)	20 A	1-pole	GFCI	12 AWG	6	—
16	Garage receptacles	20 A	1-pole	GFCI	12 AWG	8	—
17	Garage door openers	20 A	1-pole	GFCI	12 AWG	2	—
18	Exterior receptacles	20 A	1-pole	GFCI	12 AWG	4	—
19	Master suite receptacles	20 A	1-pole	AFCI	12 AWG	7	—
20	Dining and mud receptacles	20 A	1-pole	AFCI	12 AWG	6	—
21	Living receptacles	20 A	1-pole	AFCI	12 AWG	7	—
22	Office receptacles	20 A	1-pole	AFCI	12 AWG	7	—
23	Bedroom 1 receptacles	20 A	1-pole	AFCI	12 AWG	6	—
24	Bedroom 2 and hall receptacles	20 A	1-pole	AFCI	12 AWG	7	—
25	Bedroom 3 receptacles	20 A	1-pole	AFCI	12 AWG	6	—
26	Lights: master suite, laundry, mud, powder	15 A	1-pole	AFCI	14 AWG	26	—
27	Lights: kitchen, dining, pantry	15 A	1-pole	AFCI	14 AWG	19	—
28	Lights: living, entry, office, porch, patio	15 A	1-pole	AFCI	14 AWG	34	—
29	Lights: bedroom wing	15 A	1-pole	AFCI	14 AWG	30	—
30	Lights: garage	15 A	1-pole	—	14 AWG	6	—
31	Smoke / CO alarms (interconnected)	15 A	1-pole	AFCI	14 AWG	8	—

Device schedule

Tag	Device	Room	Height	Ckt
R1	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19
R2	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19
R3	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19
R4	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19
R5	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19

Tag	Device	Room	Height	Ckt
R6	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19
R7	Receptacle, tamper-resistant, AFCI	Master Bed	1'-4"	19
R8	Vanity receptacle, GFCI, 20 A bath circuit	Master Bath	3'-4"	15
R9	Vanity receptacle, GFCI, 20 A bath circuit	Master Bath	3'-4"	15
R10	Washer receptacle, 20 A dedicated	Laundry	3'-6"	12
R11	Dryer receptacle, 30 A 240 V (NEMA 14-30)	Laundry	3'-0"	13
R12	Counter receptacle, GFCI	Laundry	3'-8"	14
R13	Vanity receptacle, GFCI, 20 A bath circuit	Powder	3'-4"	15
R14	Receptacle	Mud Room	1'-4"	20
R15	Receptacle	Mud Room	1'-4"	20
R16	Receptacle, tamper-resistant, AFCI/GFCI dual function	Dining	1'-4"	20
R17	Receptacle, tamper-resistant, AFCI/GFCI dual function	Dining	1'-4"	20
R18	Receptacle, tamper-resistant, AFCI/GFCI dual function	Dining	1'-4"	20
R19	Receptacle, tamper-resistant, AFCI/GFCI dual function	Dining	1'-4"	20
R20	Counter receptacle, GFCI, 20 A	Kitchen	3'-8"	10
R21	Counter receptacle, GFCI, 20 A	Kitchen	3'-8"	10
R22	Counter receptacle, GFCI, 20 A	Kitchen	3'-8"	10
R23	Counter receptacle, GFCI, 20 A	Kitchen	3'-8"	10
R24	Island receptacle, GFCI (side of island)	Kitchen	1'-0"	11
R25	Island receptacle, GFCI (side of island)	Kitchen	1'-0"	11
R26	Counter receptacle, GFCI	Kitchen	3'-8"	11
R27	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R28	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R29	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R30	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R31	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R32	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R33	Receptacle, tamper-resistant, AFCI	Living	1'-4"	21
R34	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22
R35	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22
R36	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22
R37	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22

Tag	Device	Room	Height	Ckt
R38	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22
R39	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22
R40	Receptacle, tamper-resistant, AFCI	Office	1'-4"	22
R41	Hall receptacle (hall over 10 ft)	Bedroom Hall	1'-4"	24
R42	Receptacle, tamper-resistant, AFCI	Bedroom 1	1'-4"	23
R43	Receptacle, tamper-resistant, AFCI	Bedroom 1	1'-4"	23
R44	Receptacle, tamper-resistant, AFCI	Bedroom 1	1'-4"	23
R45	Receptacle, tamper-resistant, AFCI	Bedroom 1	1'-4"	23
R46	Receptacle, tamper-resistant, AFCI	Bedroom 1	1'-4"	23
R47	Receptacle, tamper-resistant, AFCI	Bedroom 1	1'-4"	23
R48	Vanity receptacle, GFCI, 20 A bath circuit	Bath 1	3'-4"	15
R49	Receptacle, tamper-resistant, AFCI	Bedroom 2	1'-4"	24
R50	Receptacle, tamper-resistant, AFCI	Bedroom 2	1'-4"	24
R51	Receptacle, tamper-resistant, AFCI	Bedroom 2	1'-4"	24
R52	Receptacle, tamper-resistant, AFCI	Bedroom 2	1'-4"	24
R53	Receptacle, tamper-resistant, AFCI	Bedroom 2	1'-4"	24
R54	Receptacle, tamper-resistant, AFCI	Bedroom 2	1'-4"	24
R55	Vanity receptacle, GFCI, 20 A bath circuit	Bath 2	3'-4"	15
R56	Vanity receptacle, GFCI, 20 A bath circuit	Bath 2	3'-4"	15
R57	Receptacle, tamper-resistant, AFCI	Bedroom 3	1'-4"	25
R58	Receptacle, tamper-resistant, AFCI	Bedroom 3	1'-4"	25
R59	Receptacle, tamper-resistant, AFCI	Bedroom 3	1'-4"	25
R60	Receptacle, tamper-resistant, AFCI	Bedroom 3	1'-4"	25
R61	Receptacle, tamper-resistant, AFCI	Bedroom 3	1'-4"	25
R62	Receptacle, tamper-resistant, AFCI	Bedroom 3	1'-4"	25
R63	Garage receptacle, GFCI	Garage	3'-6"	16
R64	Garage receptacle, GFCI	Garage	3'-6"	16
R65	Garage receptacle, GFCI	Garage	3'-6"	16
R66	Garage receptacle, GFCI	Garage	3'-6"	16
R67	Garage receptacle, GFCI	Garage	3'-6"	16
R68	Garage receptacle, GFCI	Garage	3'-6"	16
R69	Garage receptacle, GFCI	Garage	3'-6"	16

Tag	Device	Room	Height	Ckt
R70	Garage receptacle, GFCI	Garage	3'-6"	16
R71	Ceiling receptacle for door opener 1, GFCI	Garage	ceiling	17
R72	Ceiling receptacle for door opener 2, GFCI	Garage	ceiling	17
R73	Exterior receptacle, GFCI, weather-resistant in-use cover	Front Porch	1'-6"	18
R74	Exterior receptacle, GFCI, weather-resistant in-use cover	Front Porch	1'-6"	18
R75	Exterior receptacle, GFCI, weather-resistant in-use cover	Covered Patio	1'-6"	18
R76	Exterior receptacle, GFCI, weather-resistant in-use cover	Covered Patio	1'-6"	18
A1	48" range: 50 A 240 V range receptacle if electric (confirm fuel; a dual-fuel range needs gas + 120 V)	Kitchen	6"	1
A2	Dishwasher, 20 A (island)	Kitchen	1'-0"	2
A3	Disposal, 20 A, switched at the island	Kitchen	1'-0"	3
A4	Refrigerator, 20 A dedicated	Kitchen	4'-0"	4
A5	Range hood, 15 A (over the range)	Kitchen	6'-6"	5
A6	Air handler / furnace: dedicated circuit per the equipment picked	Mechanical	5'-0"	6
A7	Water heater: 30 A 240 V if electric or heat-pump (confirm)	Mechanical	5'-0"	7
A8	Heat pump / AC condenser, disconnect within sight, 240 V (size per equipment)	Outside	3'-0"	8
A9	Fireplace insert blower, 15 A receptacle in the surround	Living	1'-0"	9
A10	Pedestal tub: none needed unless an air or heated tub is picked	Master Bath	—	—
F1	Ceiling fan with light, fan-rated box	Master Bed	—	26
L1	Ceiling light	Master Bath	—	26
L2	Recessed wet-rated, shower	Master Bath	—	26
L3	Pendant over tub (check clearance)	Master Bath	—	26
L4	Ceiling light	Master Closet	—	26
L5	Ceiling light	Master Hall	—	26
L6	Ceiling light	Laundry	—	26
L7	Ceiling light	Powder	—	26
L8	Ceiling light	Mud Room	—	26
L9	Chandelier over the table, dimmer	Dining	—	27
L10	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L11	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L12	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L13	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L14	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27

Tag	Device	Room	Height	Ckt
L15	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L16	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L17	Recessed 6" LED, vaulted ceiling (sloped trim)	Kitchen	—	27
L18	Island pendant	Kitchen	—	27
L19	Island pendant	Kitchen	—	27
L20	Island pendant	Kitchen	—	27
L21	Under-cabinet LED strip, left wall	Kitchen	—	27
L22	Ceiling light	Pantry	—	27
L23	Ceiling light	Mechanical	—	27
L24	Recessed 6" LED, vaulted ceiling (sloped trim)	Living	—	28
L25	Recessed 6" LED, vaulted ceiling (sloped trim)	Living	—	28
L26	Recessed 6" LED, vaulted ceiling (sloped trim)	Living	—	28
L27	Recessed 6" LED, vaulted ceiling (sloped trim)	Living	—	28
L28	Recessed 6" LED, vaulted ceiling (sloped trim)	Living	—	28
L29	Recessed 6" LED, vaulted ceiling (sloped trim)	Living	—	28
F2	Ceiling fan with light, fan-rated box (vault: downrod)	Living	—	28
L30	Ceiling light	Entry	—	28
L31	Ceiling light	Coat Closet	—	28
F3	Ceiling fan with light, fan-rated box	Office	—	28
L32	Ceiling light	Bedroom Hall	—	29
L33	Ceiling light	Bedroom Hall	—	29
L34	Ceiling light	Bedroom Hall	—	29
F4	Ceiling fan with light, fan-rated box	Bedroom 1	—	29
L35	Ceiling light	Bedroom 1 Closet	—	29
L36	Ceiling light	Bath 1	—	29
L37	Recessed wet-rated, shower/tub	Bath 1	—	29
L38	Ceiling light	Closet (Bedroom 2)	—	29
F5	Ceiling fan with light, fan-rated box	Bedroom 2	—	29
L39	Ceiling light	Bath 2	—	29
L40	Recessed wet-rated, shower/tub	Bath 2	—	29
F6	Ceiling fan with light, fan-rated box	Bedroom 3	—	29
L41	Ceiling light	Closet (Bedroom 3)	—	29

Tag	Device	Room	Height	Ckt
L42	4 ft LED shop fixture	Garage	—	30
L43	4 ft LED shop fixture	Garage	—	30
L44	4 ft LED shop fixture	Garage	—	30
L45	4 ft LED shop fixture	Garage	—	30
L46	Recessed porch light, wet-rated	Front Porch	—	28
L47	Recessed porch light, wet-rated	Front Porch	—	28
L48	Recessed porch light, wet-rated	Front Porch	—	28
L49	Recessed porch light, wet-rated	Front Porch	—	28
F7	Ceiling fan with light, wet-rated, vaulted patio	Covered Patio	—	28
L50	Recessed wet-rated	Covered Patio	—	28
L51	Recessed wet-rated	Covered Patio	—	28
L52	Front door sconce, dark-sky	Outside	6'-6"	28
L53	Front door sconce, dark-sky	Outside	6'-6"	28
L54	Patio door sconce, dark-sky	Outside	6'-6"	28
L55	Patio door sconce, dark-sky	Outside	6'-6"	28
L56	Service door light, dark-sky	Outside	6'-6"	28
L57	Garage door light, dark-sky	Outside	6'-6"	28
L58	Garage door light, dark-sky	Outside	6'-6"	28
L59	Garage door light, dark-sky	Outside	6'-6"	28
F8	Exhaust fan, ducted outside, 80 CFM (humidity control on full baths)	Master Bath	—	26
F9	Exhaust fan, ducted outside, 80 CFM (humidity control on full baths)	Master Bath	—	26
F10	Exhaust fan, ducted outside, 80 CFM (humidity control on full baths)	Bath 1	—	29
F11	Exhaust fan, ducted outside, 80 CFM (humidity control on full baths)	Bath 2	—	29
F12	Exhaust fan, ducted outside, 80 CFM (humidity control on full baths)	Powder	—	26
F13	Exhaust fan, ducted outside, 80 CFM (humidity control on full baths)	Laundry	—	26
D1	Smoke alarm, interconnected, hardwired with battery backup	Bedroom 1	—	31
D2	Smoke alarm, interconnected, hardwired with battery backup	Bedroom 2	—	31
D3	Smoke alarm, interconnected, hardwired with battery backup	Bedroom 3	—	31
D4	Smoke alarm, interconnected, hardwired with battery backup	Master Bed	—	31
D5	Smoke alarm, interconnected, hardwired with battery backup	Office	—	31
D6	Combination smoke / CO alarm, interconnected, hardwired	Bedroom Hall	—	31
D7	Combination smoke / CO alarm, interconnected, hardwired	Master Hall	—	31

Tag	Device	Room	Height	Ckt
D8	Combination smoke / CO alarm, interconnected, hardwired	Living	—	31
S1	Switch at Front entry (D1): room lights, 3-way	Entry	4'-0"	28
S2	Switch at Patio French doors (D2): room lights, 3-way	Living	4'-0"	28
S3	Switch at Garage service door (D3): room lights, 3-way	Garage	4'-0"	30
S4	Switch at Garage to mud room (D6): room lights, 3-way	Garage	4'-0"	30
S5	Switch at Garage to mud room (D6): room lights, 3-way	Mud Room	4'-0"	26
S6	Switch at Master bedroom (D8): room lights, 3-way	Master Hall	4'-0"	26
S7	Switch at Master bedroom (D8): room lights	Master Bed	4'-0"	26
S8	Switch at Master bed to bath (D9): room lights	Master Bed	4'-0"	26
S9	Switch at Master bed to bath (D9): room lights	Master Bath	4'-0"	26
S10	Switch at Master bath to closet (D10): room lights	Master Bath	4'-0"	26
S11	Switch at Master closet to laundry (D11): room lights	Laundry	4'-0"	26
S12	Switch at Laundry (D12): room lights, 3-way	Mud Room	4'-0"	26
S13	Switch at Laundry (D12): room lights	Laundry	4'-0"	26
S14	Switch at Powder (D13): room lights, 3-way	Mud Room	4'-0"	26
S15	Switch at Powder (D13): room lights	Powder	4'-0"	26
S16	Switch at Pantry (D14): room lights, 3-way	Kitchen	4'-0"	27
S17	Switch at Pantry (D14): room lights	Pantry	4'-0"	27
S18	Switch at Coat closet (D15): room lights, 3-way	Entry	4'-0"	28
S19	Switch at Coat closet (D15): room lights	Coat Closet	4'-0"	28
S20	Switch at Office French doors (D16): room lights, 3-way	Entry	4'-0"	28
S21	Switch at Office French doors (D16): room lights	Office	4'-0"	28
S22	Switch at Bedroom 1 (D17): room lights, 3-way	Bedroom Hall	4'-0"	29
S23	Switch at Bedroom 1 (D17): room lights	Bedroom 1	4'-0"	29
S24	Switch at Bath 1 (D19): room lights	Bedroom 1	4'-0"	29
S25	Switch at Bath 1 (D19): room lights	Bath 1	4'-0"	29
S26	Switch at Bedroom 2 closet (D20): room lights	Bedroom 2	4'-0"	29
S27	Switch at Bedroom 2 closet (D20): room lights	Closet (Bedroom 2)	4'-0"	29
S28	Switch at Bedroom 2 (D21): room lights, 3-way	Bedroom Hall	4'-0"	29
S29	Switch at Bedroom 2 (D21): room lights	Bedroom 2	4'-0"	29
S30	Switch at Bath 2 (D22): room lights, 3-way	Bedroom Hall	4'-0"	29
S31	Switch at Bath 2 (D22): room lights	Bath 2	4'-0"	29

Tag	Device	Room	Height	Ckt
S32	Switch at Bedroom 3 (D23): room lights, 3-way	Bedroom Hall	4'-0"	29
S33	Switch at Bedroom 3 (D23): room lights	Bedroom 3	4'-0"	29
S34	Switch at Bedroom 3 closet (D24): room lights	Bedroom 3	4'-0"	29
S35	Switch at Bedroom 3 closet (D24): room lights	Closet (Bedroom 3)	4'-0"	29
S36	Switch at Kitchen to master hall (O1): room lights, 3-way	Kitchen	4'-0"	27
S37	Switch at Kitchen to master hall (O1): room lights, 3-way	Master Hall	4'-0"	26
S38	Switch at Kitchen to mud room (O2): room lights, 3-way	Kitchen	4'-0"	27
S39	Switch at Kitchen to mud room (O2): room lights, 3-way	Mud Room	4'-0"	26
S40	Switch at Living to bedroom hall (O3): room lights, 3-way	Living	4'-0"	28
S41	Switch at Living to bedroom hall (O3): room lights, 3-way	Bedroom Hall	4'-0"	29
S42	Switches at the front door: entry light, porch lights, front sconces	Entry	4'-0"	28
S43	Switches at the patio doors: patio fan/light, sconces	Living	4'-0"	28

Service load calculation (NEC 220.82)

Item	Basis	Load
General lighting & receptacles	3 VA × 3,185 sf	9,555 VA
Small-appliance circuits	2 × 1,500 VA	3,000 VA
Laundry circuit	1 × 1,500 VA	1,500 VA
Fixed appliances	range, DW, disposal, fridge, hood, water heater, dryer, blower, openers	26,200 VA
All other load, demand	first 10 kVA at 100 %, rest at 40 % (220.82(B))	22,102 VA
Heating / cooling	heat pump with 5 kW backup, at 100 % (assumed)	10,000 VA
Calculated load	32,102 VA ÷ 240 V	134 A → 200 A service

Bill of materials

Rough-order quantities from the device layout, for pricing. The electrician takes off the final list.

Item	Qty	Notes
Duplex receptacles, TR	46	15 A, plus boxes
GFCI receptacles	28	20 A, weather-resistant outside with in-use covers
Dedicated receptacles	2	washer 20 A, dryer 14-30R
Switches	43	single-pole and 3-way, dimmers in living, dining, master
Light fixtures	59	including recessed and exterior
Ceiling fans	7	fan-rated boxes

Item	Qty	Notes
Exhaust fans	6	80 CFM, ducted to outside
Smoke and smoke/CO alarms	8	hardwired, interconnected
Panel	1	200 A main breaker, 40 spaces, AFCI/GFCI breakers per schedule
NM-B 14/2 and 12/2	~4,500 ft	rough order from device count (about 22 ft per device)